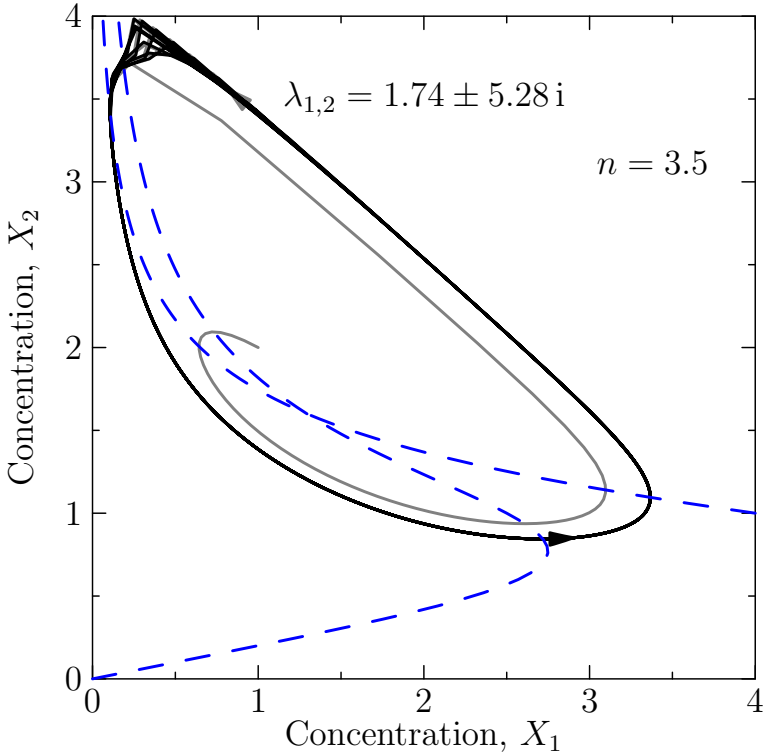
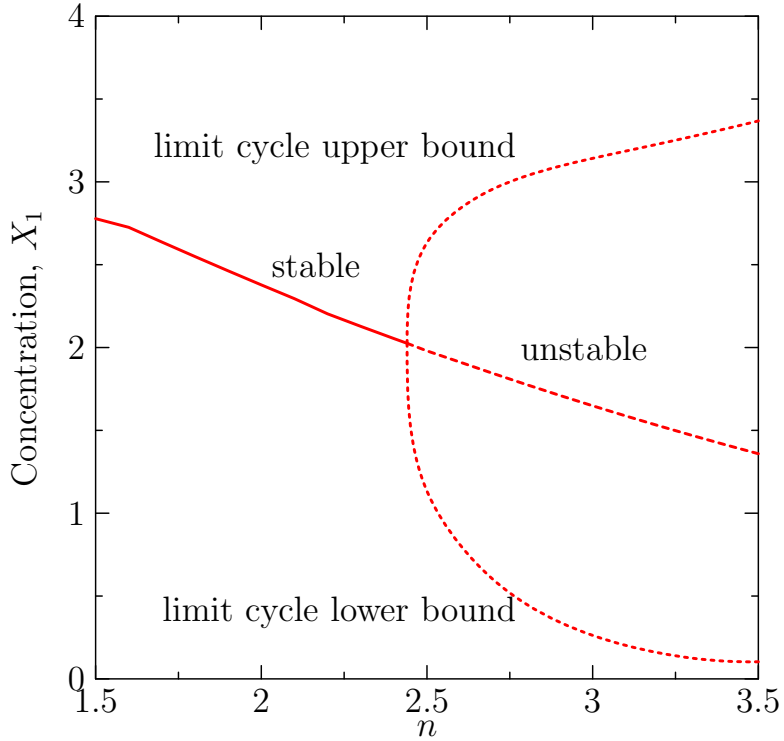
Lecture 18: Dynamical Systems

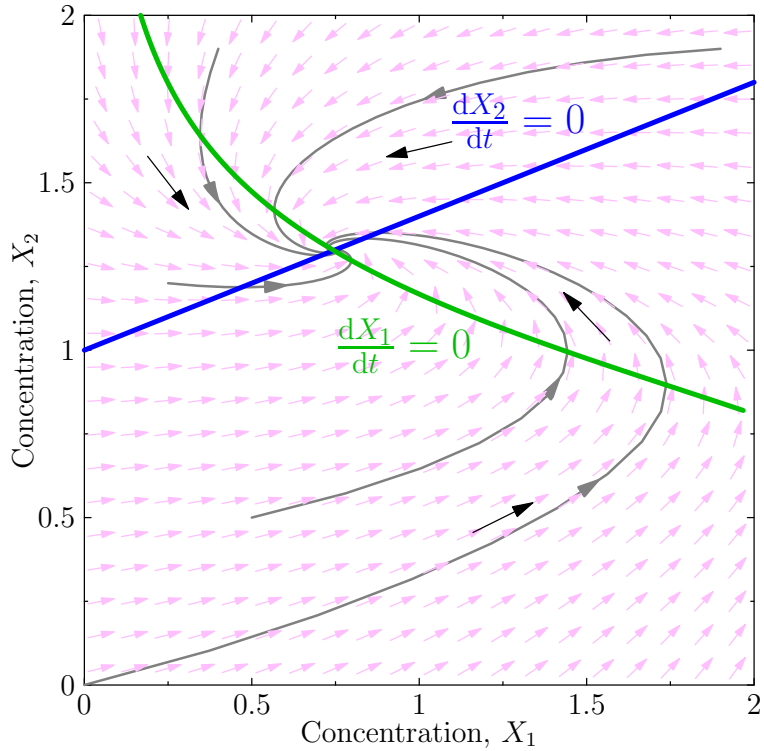




ODEs, nullclines, fixed points, stability analysis, bifications

Outline

1. **Understanding ODEs**
2. Stability
3. Bifications



Dynamical Systems

- In the last lecture we saw that we could model the dynamics of a system of molecules $\boldsymbol{X} = (X_i | i = 1, 2, \ldots, n)$ in terms of differential equations

$$\frac{\mathrm{d}X_i}{\mathrm{d}t} = f_i(\boldsymbol{X}) \qquad \text{or} \qquad \frac{\mathrm{d}[X_i]}{\mathrm{d}t} = f_i([\boldsymbol{X}])$$

- These model the mean behaviour of a set of chemical equations
- Provide the number of molecules is sufficiently large this model is accurate
- In this lecture we look at mathematical tools for understanding sets of first order differential equations

Two Species

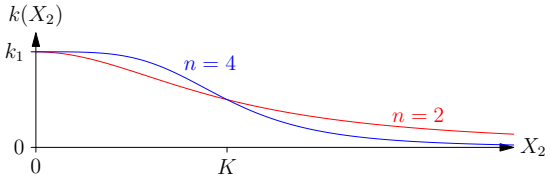
- Consider the simple system

$$\varnothing \xrightarrow{k(X_2)} X_1 \qquad X_1 \xrightarrow{k_3} \varnothing \qquad X_1 \xrightarrow{k_5} X_2$$

$$\varnothing \xrightarrow{k_2} X_2 \qquad X_2 \xrightarrow{k_4} \varnothing$$

- Where X_2 represses the expression of X_1

$$k(X_2) = \frac{k_1}{1 + (X_2/K)^n}$$

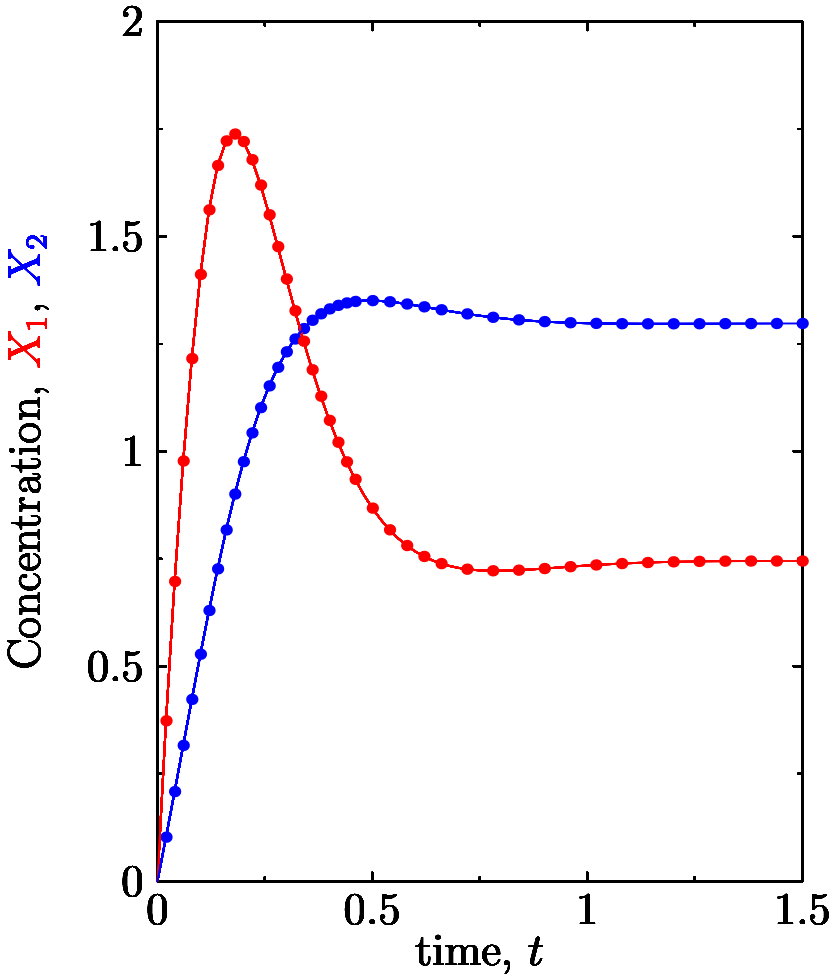


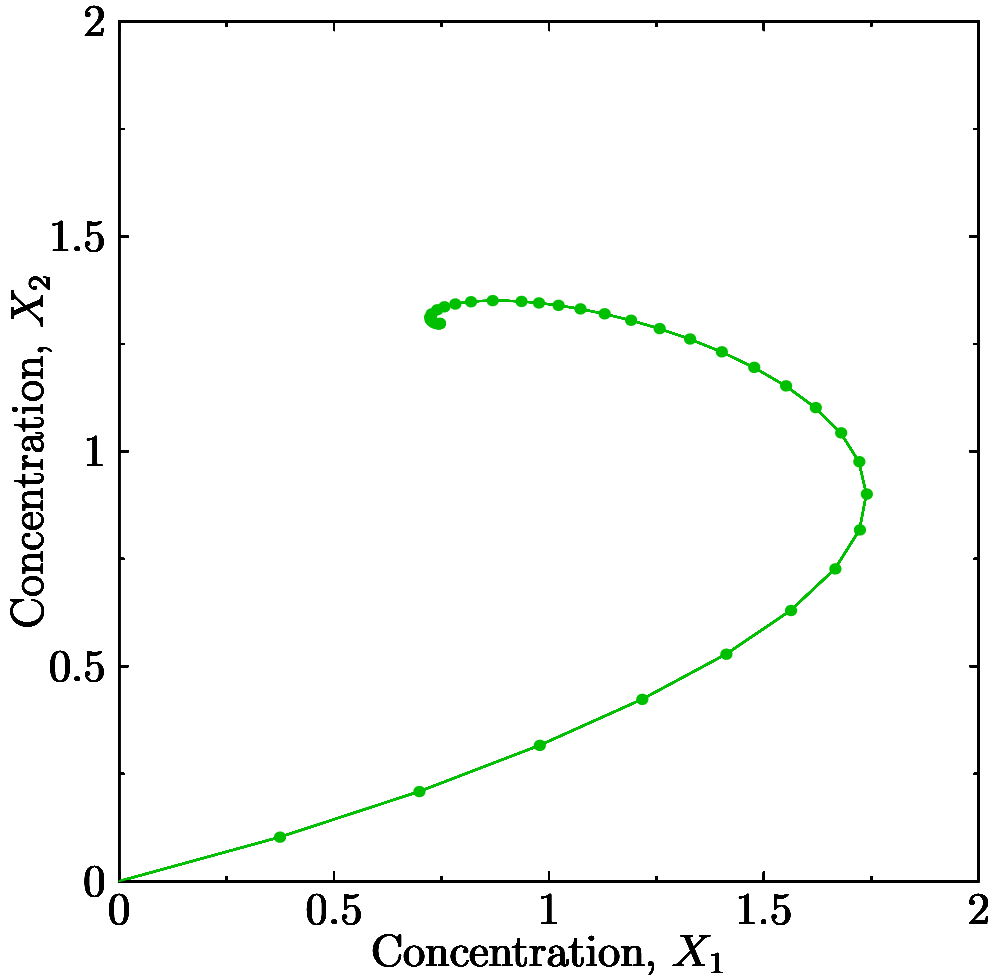
- Leading to the coupled ODEs

$$\frac{\mathrm{d}X_1(t)}{\mathrm{d}t} = f_1(X_1, X_2) = \frac{k_1}{1 + (X_2(t)/K)^n} - k_3\, X_1(t) - k_5\, X_1(t)$$

$$\frac{\mathrm{d}X_2(t)}{\mathrm{d}t} = f_2(X_1, X_2) = k_2 + k_5\, X_1(t) - k_4\, X_2(t)$$

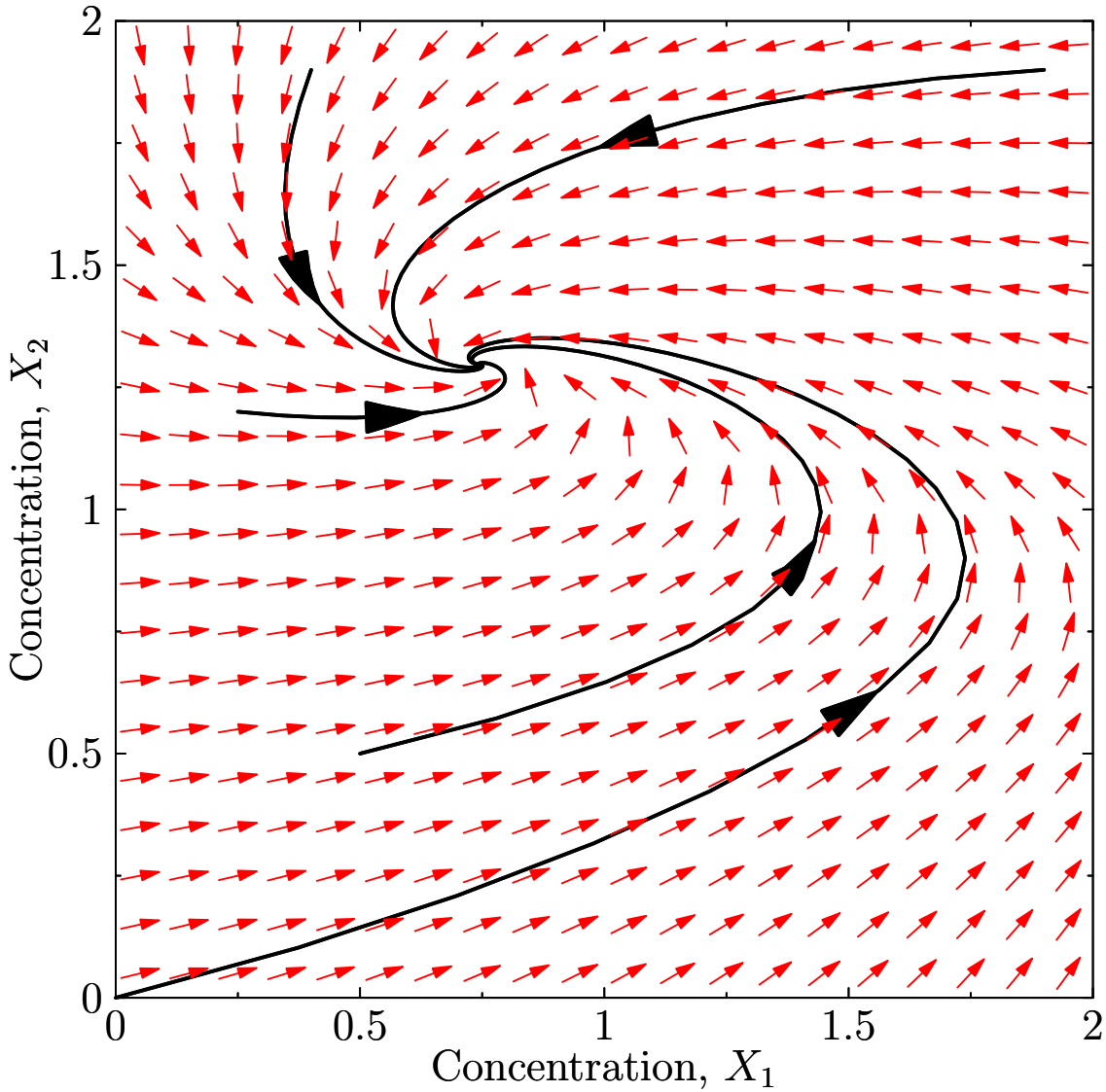
Solving the Dynamics





$$K = 1,\ k_1 = 20,\ k_2 = k_3 = k_4 = 5,\ k_5 = 2,\ n = 4$$

Trajectories



Nullclines

- To get a better understanding of the dynamics we look at the **nullclines** defined by $\mathrm{d}X_i/\mathrm{d}t = 0$

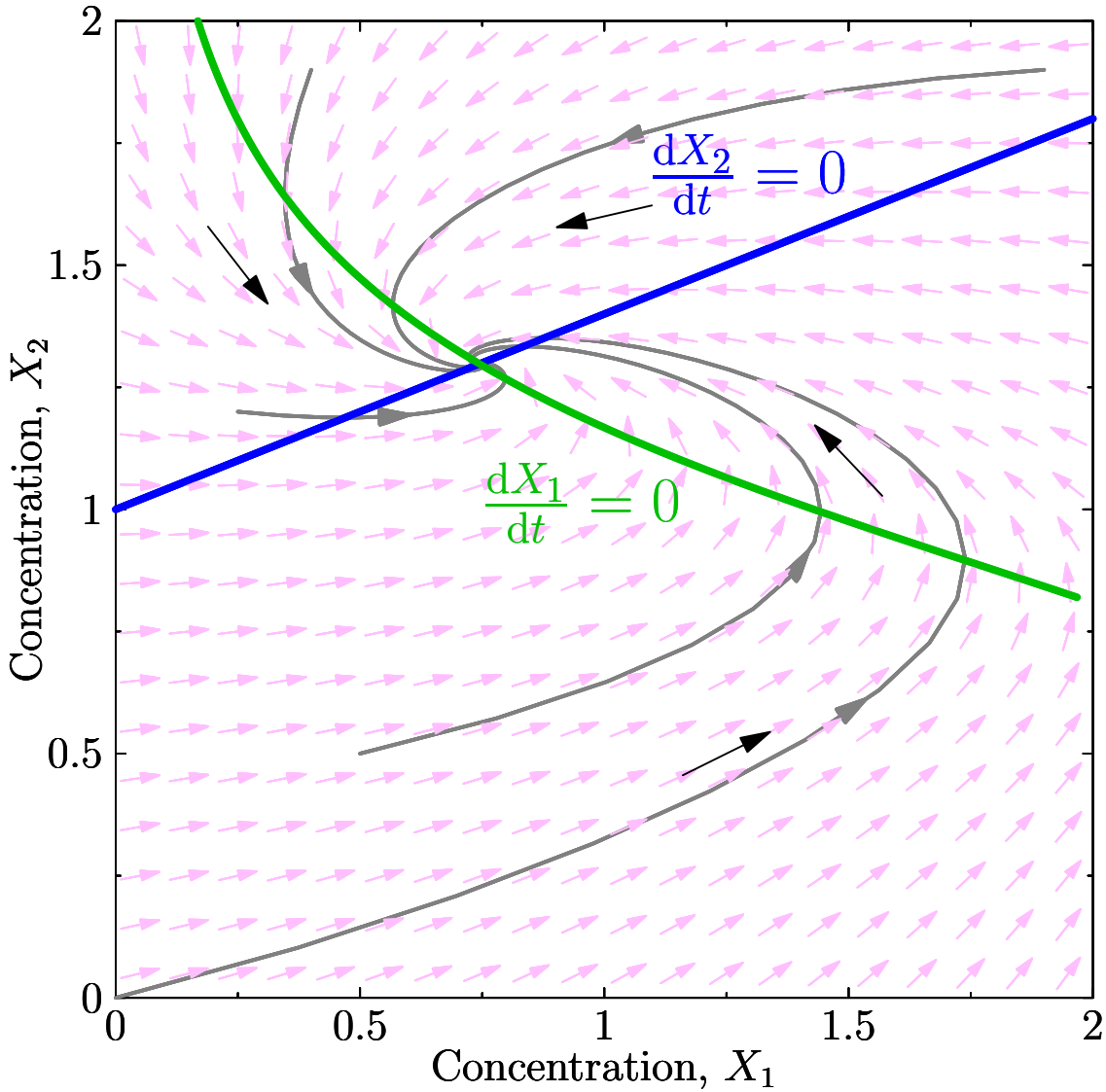
- In our system

$$\frac{\mathrm{d}X_1}{\mathrm{d}t} = \frac{k_1}{1 + (X_2/K)^n} - k_3\,X_1 - k_5\,X_1 = 0 \;\Rightarrow\; X_1 = \frac{k_1}{(k_3 + k_5)(1 + (X_2/K)^n)}$$

$$\frac{\mathrm{d}X_2}{\mathrm{d}t} = k_2 + k_5\,X_1 - k_4\,X_2 = 0 \qquad\qquad \Rightarrow\; X_2 = \frac{k_2 + k_5\,X_1}{k_2}$$

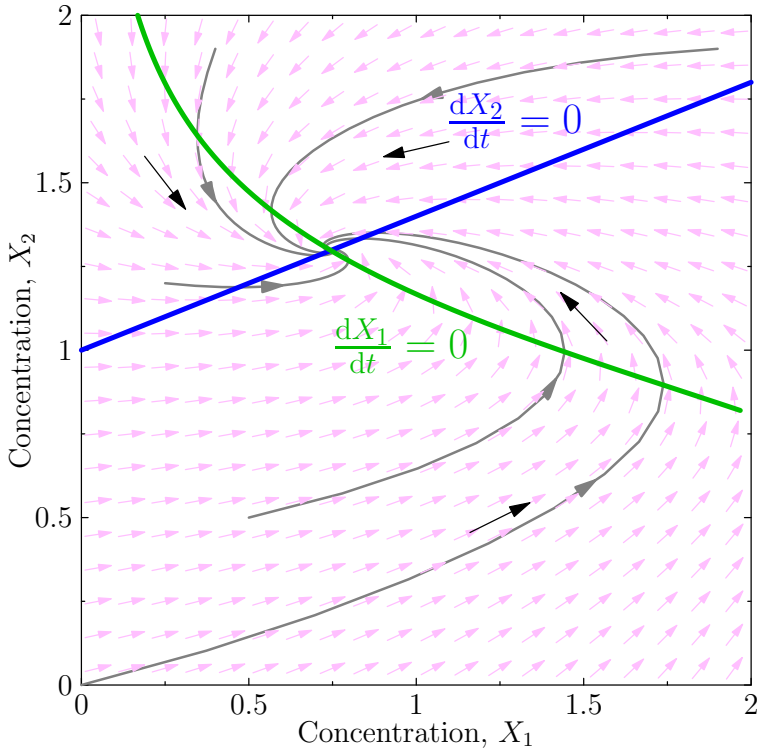
- The intersection of the nullclines defines the stationary points

Nullclines


$\frac{\mathrm{d}X_2}{\mathrm{d}t} = 0$
$\frac{\mathrm{d}X_1}{\mathrm{d}t} = 0$
Concentration, X_2
Concentration, X_1
0
0.5
1
1.5
2

Outline

1. Understanding ODEs
2. **Stability**
3. Bifications



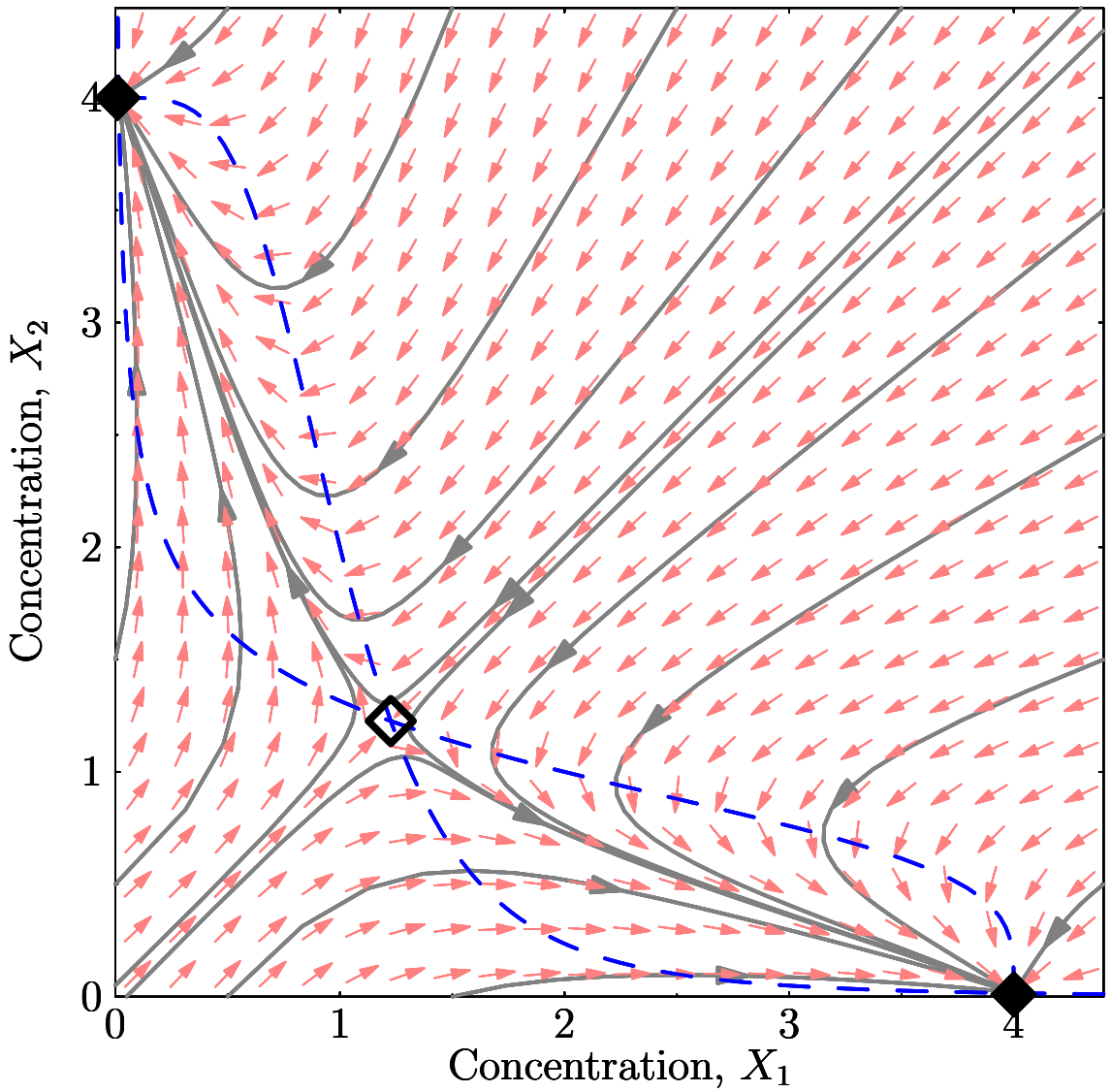
Bistability

- Let's consider a slightly different system where X_1 and X_2 inhibits each others' expression

$$\frac{\mathrm{d}X_1(t)}{\mathrm{d}t} = f_1(\boldsymbol{X}) = \frac{k_1}{1 + (X_2(t)/K_2)^{n_1}} - k_3\, X_1(t)$$
$$\frac{\mathrm{d}X_2(t)}{\mathrm{d}t} = f_2(\boldsymbol{X}) = \frac{k_2}{1 + (X_1(t)/K_1)^{n_2}} - k_4\, X_2(t)$$

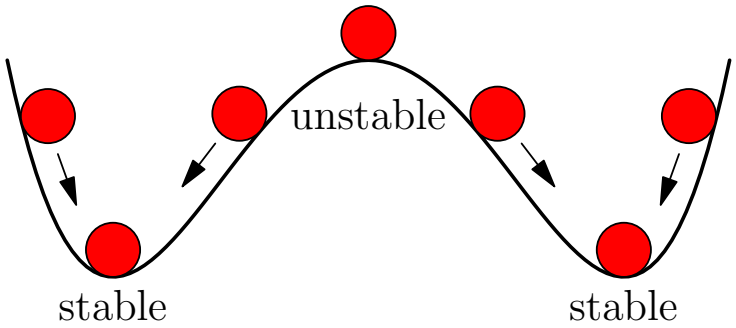
- If we consider the symmetric case $k_1 = k_2 = 20$, $K_1 = K_2 = 1$, $k_3 = k_4 = 5$, $n_1 = n_2 = 4$ we end up with three fixed points

Nullclines and Bistablity

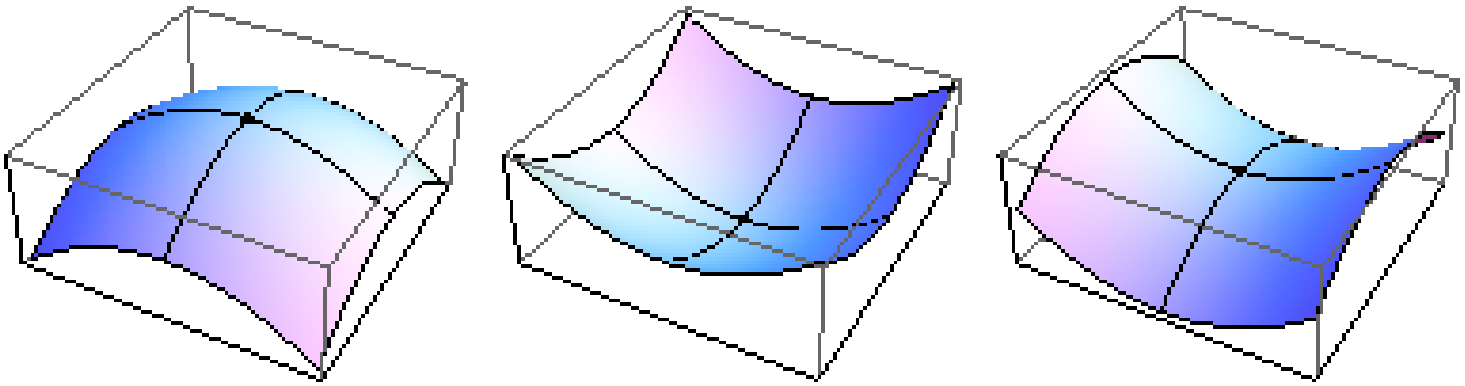


Stability

- Points can be stable or unstable



- Above 1D we can also have saddle-points



Expand Around Fixed-Point

- To study stability at a fixed-point $\boldsymbol{X}^*$ we look at what happens when we make a small perturbation $\boldsymbol{\epsilon}$

$$\frac{\mathrm{d}X_i}{\mathrm{d}t} = f_i(\boldsymbol{X}^* + \boldsymbol{\epsilon}) \approx f_i(\boldsymbol{X}^*) + \boldsymbol{\epsilon}^\mathsf{T} \boldsymbol{\nabla} f_i(\boldsymbol{X}^*) = \boldsymbol{\epsilon}^\mathsf{T} \boldsymbol{\nabla} f_i(\boldsymbol{X}^*)$$

since $f_i(\boldsymbol{X}^*) = 0$ (we are at a fixed-point)

- Where

$$\boldsymbol{\epsilon}^\mathsf{T} \boldsymbol{\nabla} f_i(\boldsymbol{X}^*) = \epsilon_1 \frac{\partial f_i(\boldsymbol{X}^*)}{\partial X_1} + \epsilon_2 \frac{\partial f_i(\boldsymbol{X}^*)}{\partial X_2} + \cdots + \epsilon_n \frac{\partial f_i(\boldsymbol{X}^*)}{\mathrm{d}X_n}$$

Jacobian

- In vector form

$$\frac{\mathrm{d}\boldsymbol{X}}{\mathrm{d}t} = \boldsymbol{f}(\boldsymbol{X}^* + \boldsymbol{\epsilon}) = \mathbf{J}\,\boldsymbol{\epsilon} + O(\boldsymbol{\epsilon}^2)$$

- where **J** is the **Jacobian** matrix with elements

$$J_{ij} = \left.\frac{\partial f_i(\boldsymbol{X})}{\partial X_j}\right|_{\boldsymbol{X}=\boldsymbol{X}^*}$$

- Note that this is just saying

$$\frac{\mathrm{d}X_i}{\mathrm{d}t} \approx \epsilon_1 \frac{\partial f_i(\boldsymbol{X}^*)}{\partial X_1} + \epsilon_2 \frac{\partial f_i(\boldsymbol{X}^*)}{\partial X_2} + \cdots + \epsilon_n \frac{\partial f_i(\boldsymbol{X}^*)}{\mathrm{d}X_n}$$

Linear Approximation

- Let $\boldsymbol{\epsilon}(t) = \boldsymbol{X}(t) - \boldsymbol{X}^*$ then (since $\mathrm{d}\boldsymbol{\epsilon}(t) = \mathrm{d}\boldsymbol{X}(t)$)

$$\frac{\mathrm{d}\boldsymbol{\epsilon}(t)}{\mathrm{d}t} = \frac{\mathrm{d}\boldsymbol{X}(t)}{\mathrm{d}t} \approx \mathbf{J}\,\boldsymbol{\epsilon}(t)$$

- To solve this we note that we can write most square matrices $\mathbf{J}$ as

$$\mathbf{J} = \mathbf{V}\boldsymbol{\Lambda}\,\mathbf{V}^{-1}$$

- Where $\boldsymbol{\Lambda}$ is a diagonal matrix whose elements $\Lambda_{ii} = \lambda_i$ are eigenvalues of $\mathbf{J}$ and the columns of $\mathbf{V}$ are the eigenvectors

Eigensystem of a General Matrix

- For a general (non-symmetric) matrix $\mathbf{M}$ the vector $\boldsymbol{v}_i$ is an (right-handed) eigenvector if

$$\mathbf{M}\,\boldsymbol{v}_i = \lambda_i\,\boldsymbol{v}_i$$

- There are also left-handed eigenvectors $\boldsymbol{u}_i$ with the same eigenvalues, λ_i, such that

$$\boldsymbol{u}_i^\mathsf{T}\,\mathbf{M} = \lambda_i\,\boldsymbol{u}_i^\mathsf{T}$$

- In general $\boldsymbol{v}_i \neq \boldsymbol{u}_i$ although if $\lambda_i \neq \lambda_j$ then $\boldsymbol{u}_i^\mathsf{T}\boldsymbol{v}_j = 0$

- The eigenvalues and eigen-vectors are either real or exist in complex conjugate pairs $\lambda_j = \lambda_i^*$ and $\boldsymbol{v}_j = \boldsymbol{v}_i^*$

Matrix Decomposition

- If we write

$$\mathbf{V} = \begin{pmatrix} \boldsymbol{v}_1 & \boldsymbol{v}_2 & \cdots & \boldsymbol{v}_n \end{pmatrix} \qquad \boldsymbol{\Lambda} = \begin{pmatrix} \lambda_1 & 0 & \cdots & 0 \\ 0 & \lambda_2, & \cdots & 0 \\ \vdots & \vdots & \ddots & 0 \\ 0 & 0 & \cdots & \lambda_n \end{pmatrix}$$

- Then $\mathbf{M} = \mathbf{V}\,\boldsymbol{\Lambda}\,\mathbf{V}^{-1} = \mathbf{V}\,\boldsymbol{\Lambda}\,\mathbf{U} = \mathbf{U}^{-1}\,\boldsymbol{\Lambda}\,\mathbf{U}$

- where $\mathbf{U} = \mathbf{V}^{-1}$ has rows equal to $\boldsymbol{u}_i^{\mathsf{T}}$

- Multiplying $\mathbf{M}$ on the right by $\mathbf{V}$ we find

$$\mathbf{M}\,\mathbf{V} = (\mathbf{V}\,\boldsymbol{\Lambda}\,\mathbf{V}^{-1})\,\mathbf{V} = \mathbf{V}\,\boldsymbol{\Lambda} \qquad \Rightarrow \qquad \mathbf{M}\,\boldsymbol{v}_i = \lambda_i\,\boldsymbol{v}_i$$

- Multiplying $\mathbf{M}$ on the left by $\mathbf{U} = \mathbf{V}^{-1}$

$$\mathbf{U}\,\mathbf{M} = \mathbf{U}\,(\mathbf{U}^{-1}\,\boldsymbol{\Lambda}\,\mathbf{U}) = \boldsymbol{\Lambda}\,\mathbf{U} \qquad \Rightarrow \qquad \boldsymbol{u}_i^{\mathsf{T}}\,\mathbf{M} = \lambda_i\,\boldsymbol{u}_i^{\mathsf{T}}$$

Linearised Solution

- Multiply the linear approximation by $\mathbf{V}^{-1}$

$$\frac{\mathrm{d}\boldsymbol{\epsilon}(t)}{\mathrm{d}t} = \mathbf{V}\boldsymbol{\Lambda}\,\mathbf{V}^{-1}\,\boldsymbol{\epsilon}(t) \qquad \Rightarrow \qquad \frac{\mathrm{d}\mathbf{V}^{-1}\boldsymbol{\epsilon}(t)}{\mathrm{d}t} = \boldsymbol{\Lambda}\,\mathbf{V}^{-1}\boldsymbol{\epsilon}(t)$$

- Writing $\hat{\boldsymbol{\epsilon}}(t) = \mathbf{V}^{-1}\boldsymbol{\epsilon}(t)$ then

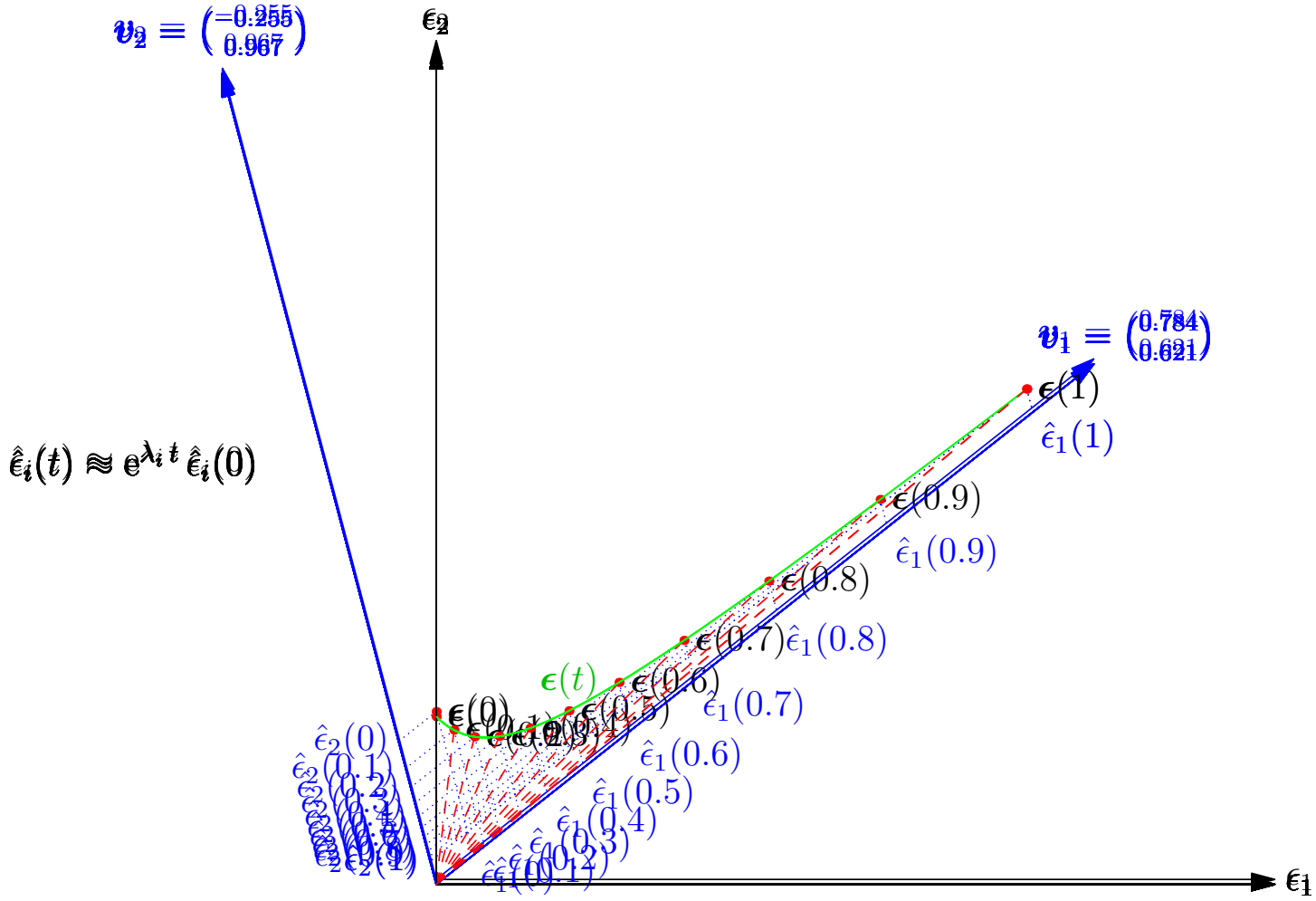
$$\frac{\mathrm{d}\hat{\boldsymbol{\epsilon}}(t)}{\mathrm{d}t} = \boldsymbol{\Lambda}\,\hat{\boldsymbol{\epsilon}}(t) \qquad \text{or} \qquad \frac{\mathrm{d}\hat{\epsilon}_i(t)}{\mathrm{d}t} = \lambda_i\,\hat{\epsilon}_i(t)$$

- With solution

$$\hat{\epsilon}_i(t) = \mathrm{e}^{\lambda_i\,t}\,\hat{\epsilon}_i(0)$$

Example

$$\mathbf{J} = \begin{pmatrix} 2 & 1 \\ 3 & -1 \end{pmatrix} = \mathbf{V}\boldsymbol{\Lambda}\mathbf{V}^{-1} = \begin{pmatrix} 0.784 & -0.255 \\ 0.621 & 0.967 \end{pmatrix} \begin{pmatrix} 2.791 & 0 \\ 0 & -1.791 \end{pmatrix} \begin{pmatrix} 1.055 & 0.278 \\ -0.677 & 0.856 \end{pmatrix}$$



Stability Analysis

- If $\Re(\lambda_i) < 0$ for all i the fixed point is stable (all sufficiently small perturbations shrink)
- If there exists an i such that $\Re(\lambda_i) > 0$ then the fixed point is unstable
- If $\Re(\lambda_i) < 0$ and $\Re(\lambda_j) > 0$ then we are at an unstable saddle point
- Note that the Jacobian is not symmetric which means the eigenvalues can occur in complex conjugate pairs (i.e. $\lambda_i^* = \lambda_j$)
- In this case the trajectories will either spiral in to the fixed point or spiral away from the fixed point

Oscillator

- Consider a chain with a positive feedback

$$\varnothing \xrightarrow{k_0} X_1$$

$$X_1 \xrightarrow{k(X_2)} X_2$$

$$X_2 \xrightarrow{k_2} \varnothing$$

$$k(X_2) = k_1 \left(1 + (X_2/K)^n\right)$$

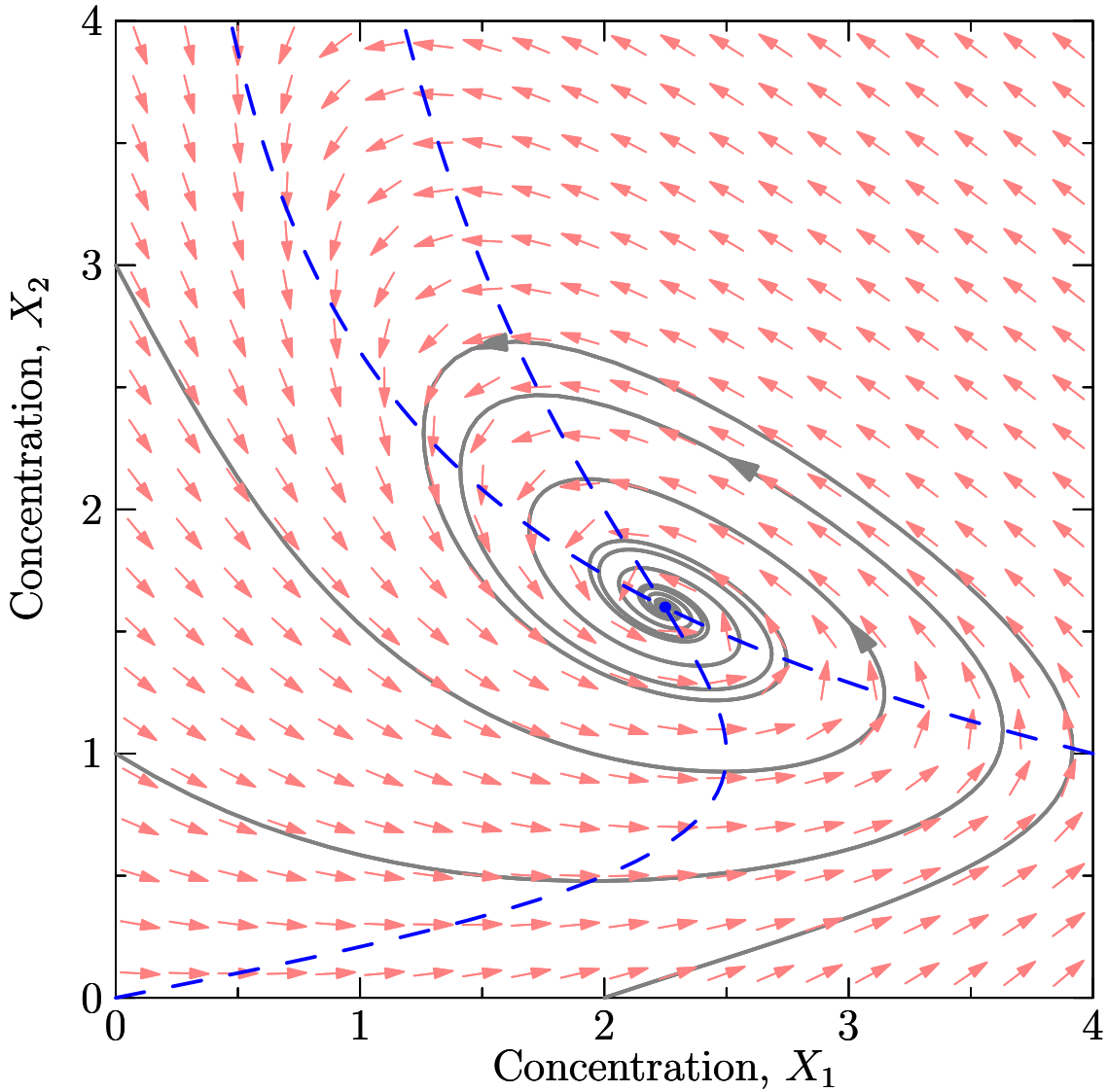
$$\frac{\mathrm{d}X_1(t)}{\mathrm{d}t} = k_0 - k_1 \left(1 + \left(\frac{X_2(t)}{K}\right)^n\right) X_1(t)$$

$$\frac{\mathrm{d}X_2(t)}{\mathrm{d}t} = k_1 \left(1 + \left(\frac{X_2(t)}{K}\right)^n\right) X_1(t) - k_2 X_2(t)$$

- Consider $k_0 = 8$, $k_1 = 1$, $K = 1$, $k_2 = 5$ and $n = 2$

- Large values of X_2 will result in most of X_1 being turned into X_2, then X_2 will decay rapidly allowing X_1 to increase

Nullclines and Bistablity



Fixed Point Equations

- Our dynamical equations are

$$\frac{\mathrm{d}X_1(t)}{\mathrm{d}t} = f_1(\boldsymbol{X}(t)) = k_0 - k_1 \left(1 + \left(\frac{X_2(t)}{K}\right)^n\right) X_1(t)$$

$$\frac{\mathrm{d}X_2(t)}{\mathrm{d}t} = f_2(\boldsymbol{X}(t)) = k_1 \left(1 + \left(\frac{X_2(t)}{K}\right)^n\right) X_1(t) - k_2\, X_2(t)$$

- With $k_0 = 8$, $k_1 = 1$, $K = 1$, $k_2 = 5$ and $n = 2$

- The fixed point is at $\boldsymbol{X}^* = \left(\frac{k_0}{k_1(1+(k_0/(K\,k_2))^n)}, \frac{k_0}{k_2}\right) = \left(\frac{200}{89}, \frac{8}{5}\right)$

- The Jacobian $J_{ij} = \partial f_i(\boldsymbol{X}(t))/\partial X_j$ is

$$\mathbf{J} = \begin{pmatrix} -k_1 \left(1 + \left(\frac{X_2^*}{K}\right)^n\right) & -\frac{n\,k_1}{K} \left(\frac{X_2^*}{K}\right)^{n-1} X_1^* \\ k_1 \left(1 + \left(\frac{X_2^*}{K}\right)^n\right) & \frac{n\,k_1}{K} \left(\frac{X_2^*}{K}\right)^{n-1} X_1^* - k_2 \end{pmatrix} = \begin{pmatrix} \frac{-89}{25} & \frac{-640}{89} \\ \frac{89}{25} & \frac{195}{89} \end{pmatrix}$$

Eigen Decompostion

- Plugging this into Matlab (using `eig(J)`)

$$\mathbf{J} = \begin{pmatrix} \frac{-89}{25} & \frac{-640}{89} \\ \frac{89}{25} & \frac{195}{89} \end{pmatrix} = \mathbf{V} \begin{pmatrix} -0.68 + 4.16\mathrm{i} & 0 \\ 0 & -0.68 - 4.16\mathrm{i} \end{pmatrix} \mathbf{V}^{-1}$$

- where

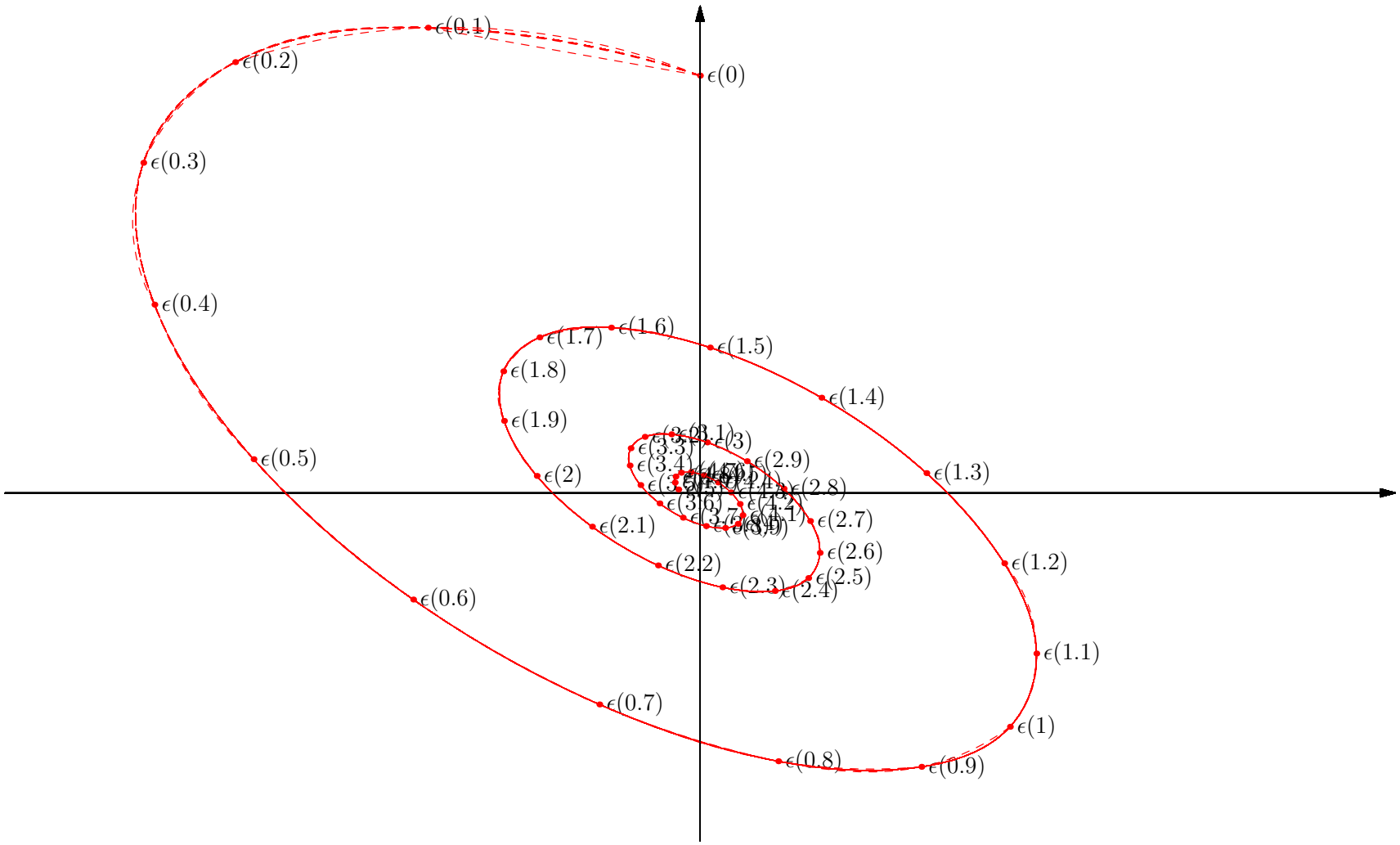
$$\mathbf{V} = \begin{pmatrix} 0.82 & 0.82 \\ -0.33 - 0.47\mathrm{i} & -0.33 + 0.47\mathrm{i} \end{pmatrix}$$

- Or $T = \mathrm{tr}\,\mathbf{J} = J_{11} + J_{22}$ and $D = \det(\mathbf{J}) = J_{11}\,J_{22} - J_{12}\,J_{21}$ then $\lambda_{1,2} = T/2 \pm \sqrt{T^2/4 - D}$

- The eigenvalues have a large complex component which causes a significant spiralling

Dynamics

$$\mathbf{J} = \begin{pmatrix} \frac{-89}{25} & \frac{-640}{89} \\ \frac{89}{25} & \frac{195}{89} \end{pmatrix} = \mathbf{V}\Lambda\mathbf{V}^{-1} = \begin{pmatrix} 0.818 & 0.818 \\ -0.327 - 0.473\mathrm{i} & -0.327 + 0.473\mathrm{i} \end{pmatrix} \begin{pmatrix} -0.685 + 4.163\mathrm{i} & 0 \\ 0 & -0.685 - 4.163\mathrm{i} \end{pmatrix} \begin{pmatrix} 0.611 + 0.422\mathrm{i} & 1.056\mathrm{i} \\ 0.611 - 0.422\mathrm{i} & -1.056\mathrm{i} \end{pmatrix}$$



Incidentally. . .

- A property of traces ($\mathrm{tr}\,\mathbf{M} = \sum_i M_{ii}$) is

$$\mathrm{tr}\,\mathbf{A}\,\mathbf{B} = \sum_{i,j} A_{ij}\,B_{ji} = \sum_{i,j} B_{ji}\,A_{ij} = \mathrm{tr}\,\mathbf{B}\,\mathbf{A}$$

- Using $\mathbf{J} = \mathbf{V}\,\boldsymbol{\Lambda}\mathbf{V}^{-1}$

$$\mathrm{tr}\,\mathbf{J} = \mathrm{tr}\,\mathbf{V}\,\boldsymbol{\Lambda}\,\mathbf{V}^{-1} = \mathrm{tr}\,\mathbf{V}^{-1}\,\mathbf{V}\,\boldsymbol{\Lambda} = \mathrm{tr}\,\boldsymbol{\Lambda} = \sum_i \lambda_i$$

- Two properties of determinants is that $|\mathbf{A}\,\mathbf{B}| = |\mathbf{A}|\,|\mathbf{B}|$ and $|\mathbf{M}^{-1}| = |\mathbf{M}|^{-1}$ thus

$$|\mathbf{J}| = |\mathbf{V}|\,|\boldsymbol{\Lambda}|\,|\mathbf{V}^{-1}| = |\boldsymbol{\Lambda}| = \prod_i \lambda_i$$

- If $\lambda_1 = T/2 - \sqrt{T^2/4 - D}$ and $\lambda_2 = T/2 + \sqrt{T^2/4 - D}$ then

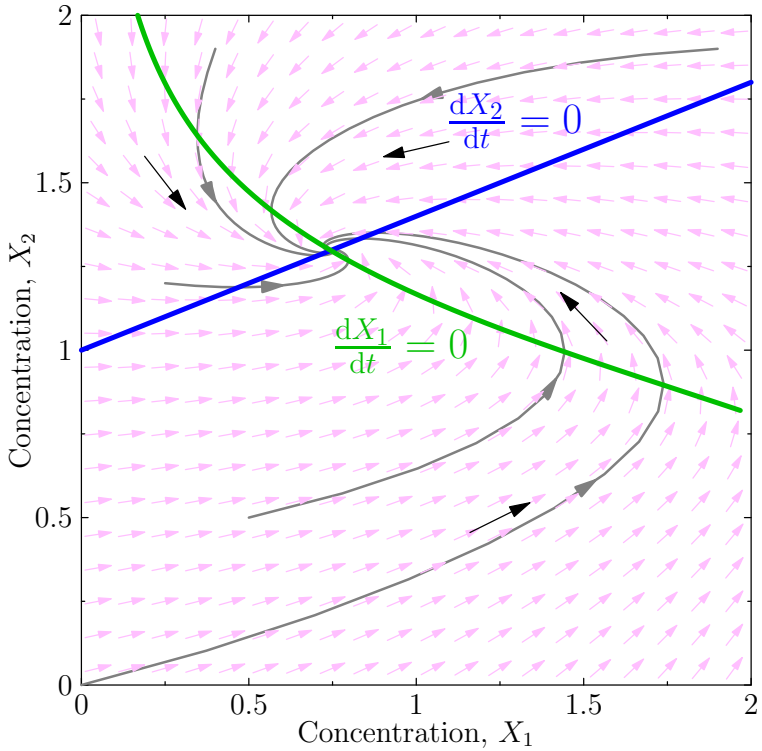
$$\lambda_1 + \lambda_2 = T \qquad\qquad \lambda_1 \times \lambda_2 = D$$

High Dimensional Systems

- We have only considered simple systems with two species of molecules
- Much of the analysis techniques generalise to higher dimensions, but it is far harder to get closed from solutions
- Using programs like Mathematica it is often easy to find fixed point solutions numerically
- However, in biology different reactions happen at very different rates
- We often concentrate on the slowest reactions and assume all the other reactions have reached equilibrium (however, beware this needs to be analysed)

Outline

1. Understanding ODEs
2. Stability
3. **Bifications**



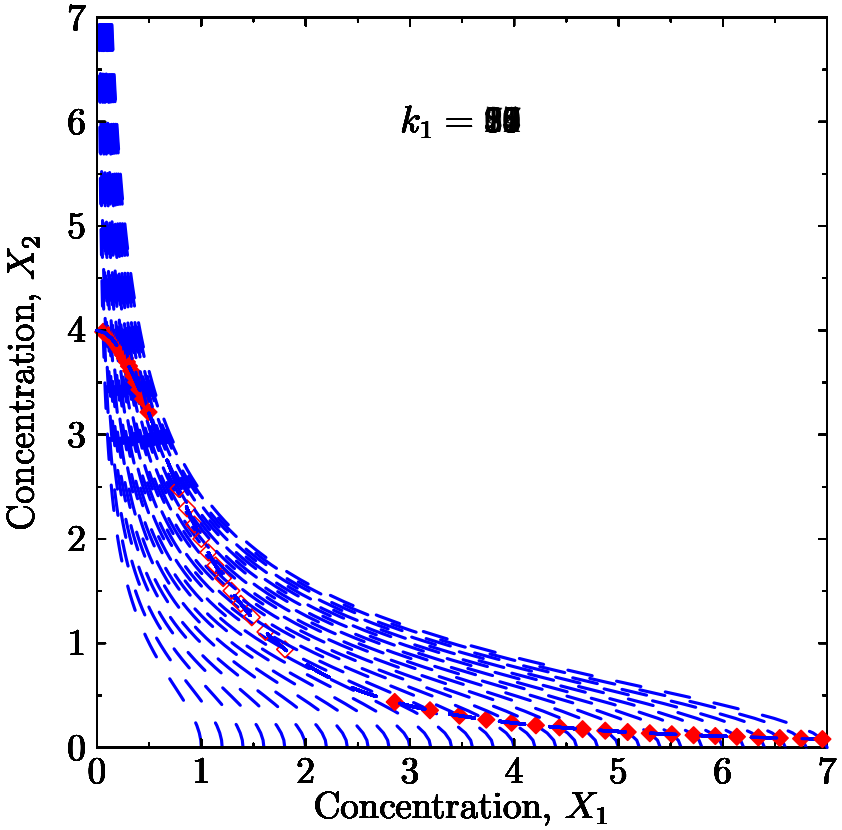
Back to Bistability

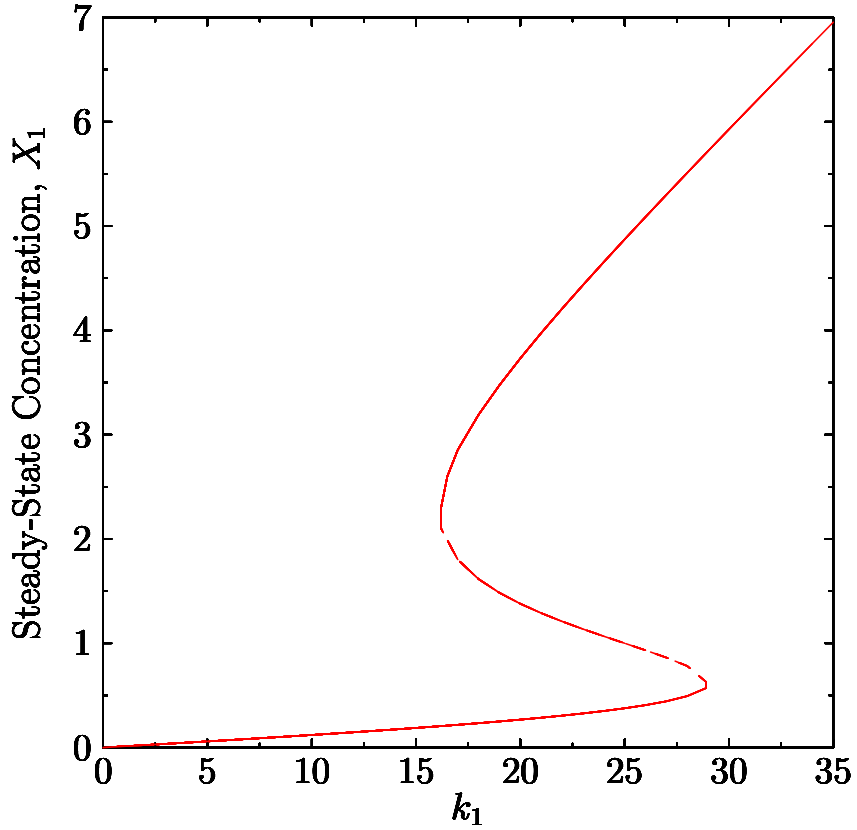
- Recall that we could create a bistable system through mutual inhibition

$$\frac{\mathrm{d}X_1(t)}{\mathrm{d}t} = \frac{k_1}{1 + (X_2(t)/K_2)^{n_1}} - k_3\, X_1(t)$$
$$\frac{\mathrm{d}X_2(t)}{\mathrm{d}t} = \frac{k_2}{1 + (X_1(t)/K_1)^{n_2}} - k_4\, X_2(t)$$

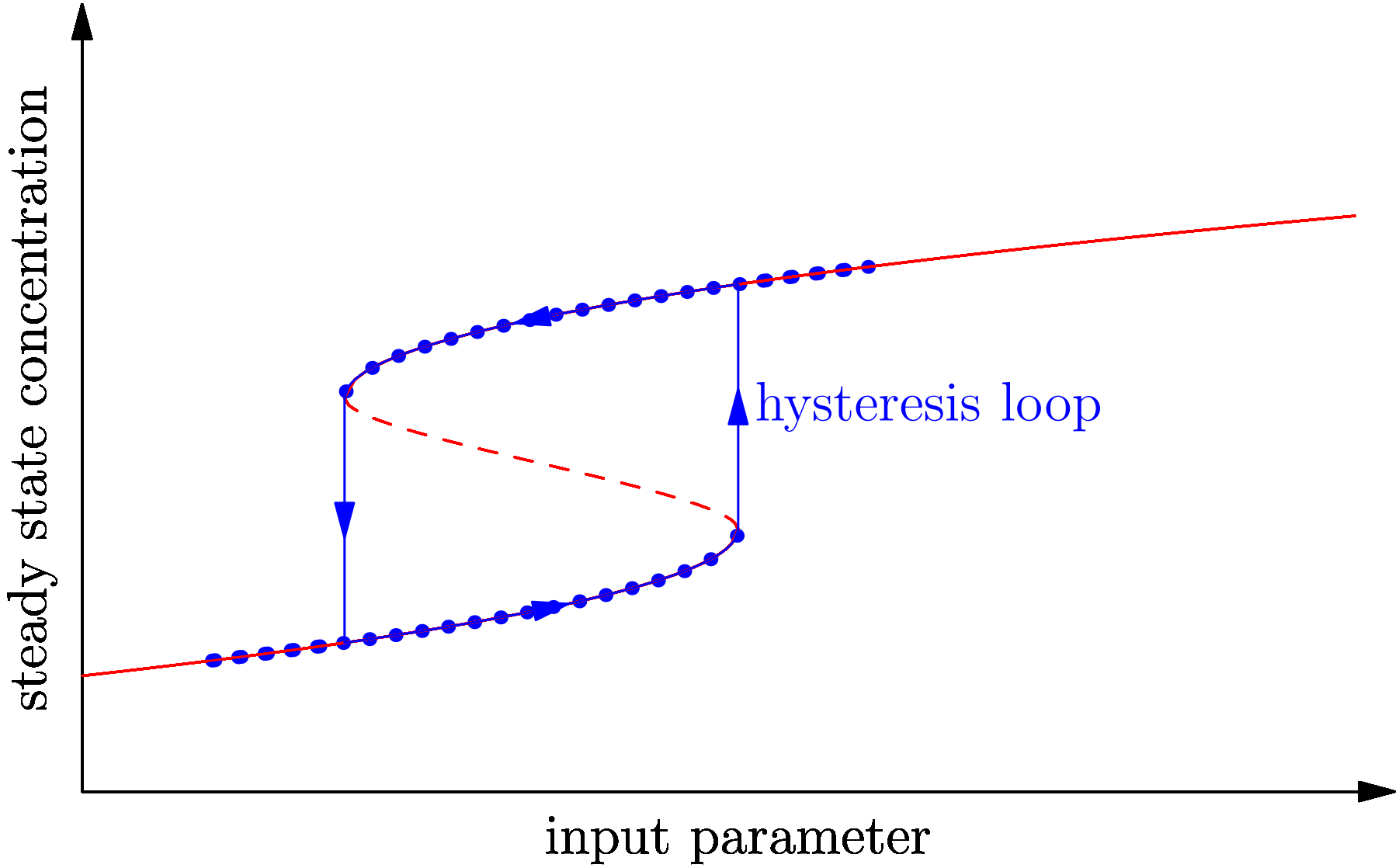
- Let $k_2 = 20$, $K_1 = K_2 = 1$, $k_3 = k_4 = 5$, $n_1 = n_2 = 2$
- But now we look at varying the parameter k_1
- The fixed points are given where the nullclines intersect

Varying k_1


$k_1 =$
Concentration, X_2
Concentration, X_1
0
1
2
3
4
5
6
7


Steady-State Concentration, X_1
k_1
0
5
10
15
20
25
30
35

Hysteresis


steady state concentration
hysteresis loop
input parameter

Autocatalysis

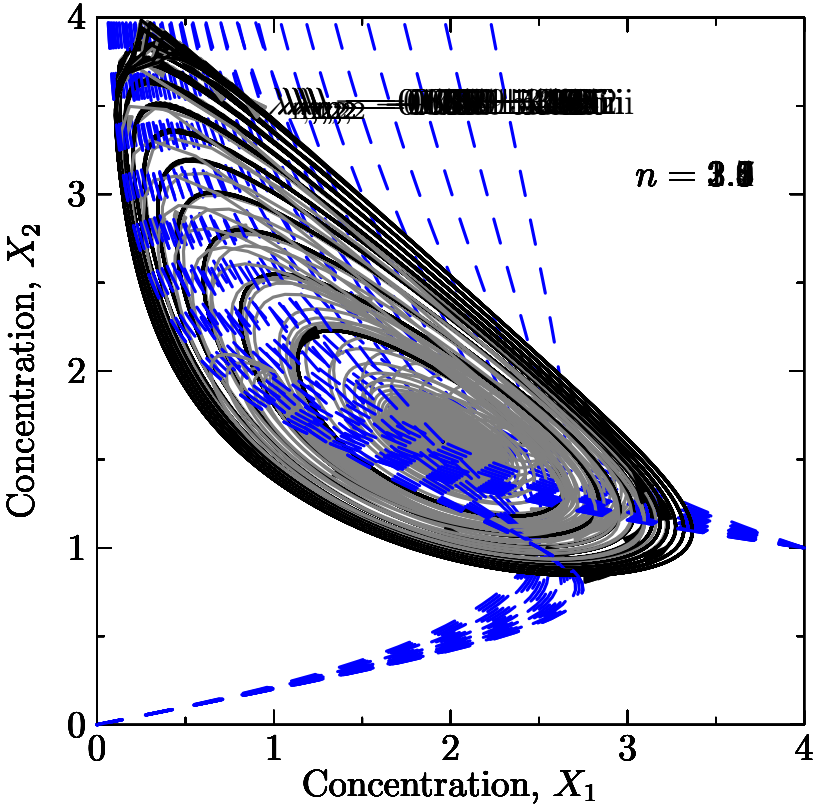
- Recall the autocatalytic system

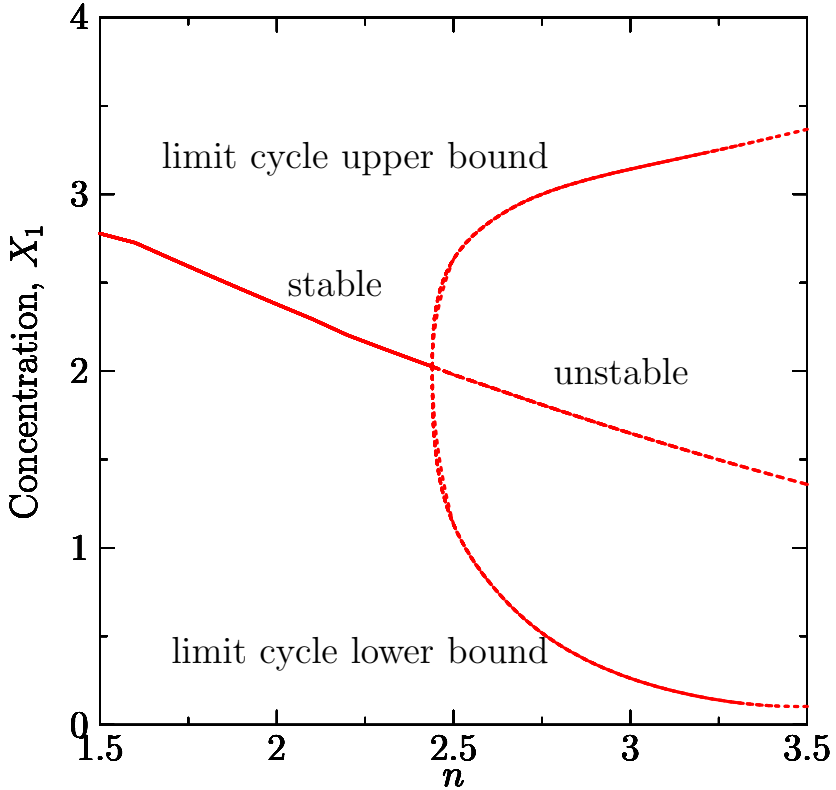
$$\frac{\mathrm{d}X_1(t)}{\mathrm{d}t} = k_0 - k_1 \left(1 + \left(\frac{X_2(t)}{K}\right)^n\right) X_1(t)$$
$$\frac{\mathrm{d}X_2(t)}{\mathrm{d}t} = k_1 \left(1 + \left(\frac{X_2(t)}{K}\right)^n\right) X_1(t) - k_2 X_2(t)$$

- With $k_0 = 8$, $k_1 = 1$, $K = 1$, $k_2 = 5$

- We consider what happens when we vary n

Hopf Bifurcation





Summary

- We can understand a lot of behaviours by understanding first order ODEs
- Systems can exhibit fixed points, limit cycles (and even chaotic behaviour in enough dimensions)
- We can study the stability of fixed points
- The behaviour of systems can change radically as new fixed points emerge or fixed points become unstable